Part 3: Gradient Descent Zoky ①

We have the following as initial data, target, weights, and biases

$$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \quad y = \begin{bmatrix} 5 \\ 6 \end{bmatrix} \quad m^{(0)} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \quad b^{(0)} = 1$$

Let $J(m, b)$ be our MSE function such that:

$$J(m, b) = \frac{1}{M} \sum_{i=1}^{M} (\hat{y}_i - y_i)^2 \text{ where } \hat{y}_i = m_1 x_{i1} + m_2 x_{i2} + b$$

Let e_i be the error of sample i such that:

$$e_i = \hat{y}_i - y_i \text{ which gives us } J = \frac{1}{M} \sum_{i=1}^{M} e_i^2$$

* Derive the cost function:

We will compute 3 partial derivatives since m has two components m_1, m_2 and b is a scalar.

- Step 1: compute $\frac{\partial J}{\partial e_i}$

Since J is a sum, differentiating with respect to a single e_i removes every other term except the i-th term (power rule)

$$\frac{\partial J}{\partial e_i} = \frac{2}{M} e_i$$

= Step 2: computing $(e_j)'$ with respect to m_1, m_2, and b: ②

$e_j = m_1 x_{j1} + m_2 x_{j2} + b - y_j$

$\frac{\partial e_j}{\partial m_1} = x_{j1}$ (coefficient of m_1 when m_1 is the variable)

$\frac{\partial e_j}{\partial m_2} = x_{j2}$ (coefficient of m_2 when m_2 is the variable)

$\frac{\partial e_j}{\partial b} = 1$ (b is a constant term in e_j)

- Step 3: applying the chain rule

A parameter affects J through every e_j, so by using the chain rule we add up all the paths:

$$\frac{\partial J}{\partial m_1} = \sum_{j=1}^{M} \frac{\partial J}{\partial e_j} \cdot \frac{\partial e_j}{m_1}$$

$$= \sum_{j=1}^{M} \frac{2}{M} e_j \cdot x_{j1}$$

$$\frac{\partial J}{\partial m_1} = \frac{2}{M} \sum_{j=1}^{M} e_j \cdot x_{j1}$$

m_2 follows similar steps to m_1, we can replace m_1, x_{j1} with m_2, x_{j2} respectively:

$$\frac{\partial J}{\partial m_2} = \frac{2}{M} \sum_{j=1}^{M} e_j \cdot x_{j2}$$

We use the chain rule for b :

$$\frac{\partial J}{\partial b} = \sum_{j=1}^{M} \frac{\partial J}{\partial e_j} \cdot \frac{\partial e_j}{\partial b}$$

$$= \sum_{j=1}^{M} \frac{2}{M} e_j \cdot 1$$

$$\frac{\partial J}{\partial b} = \frac{2}{M} \sum_{j=1}^{M} e_j$$

- Step 4 : Gradient and matrix form

We can combine the two partial derivatives into a gradient which depends on m :

$$\nabla_m J = \begin{bmatrix} \frac{\partial J}{\partial m_1} \\ \frac{\partial J}{\partial m_2} \end{bmatrix} = \frac{2}{M} \sum_{j=1}^{M} e_j \begin{bmatrix} x_{j1} \\ x_{j2} \end{bmatrix}$$

By using the column-row method described by Strang (2016, Section 2.4), we can express :

$$X^T e = \sum_{j=1}^{M} e_j \begin{bmatrix} x_{j1} \\ x_{j2} \end{bmatrix} \quad \text{therefore} \quad \nabla_m J = \frac{2}{M} X^T e$$

* Gradient descent:

We set the following rules for our computations:

$\gamma = 0.01$, $m^{(k+1)} = m^{(k)} - \gamma \nabla_m J$, $b^{(k+1)} = b^{(k)} - \gamma \frac{\partial J}{\partial b}$

(learning rate)

- iteration 0:

$$e^{(0)} = X m^{(0)} + b^{(0)} \begin{bmatrix} 1 \\ 1 \end{bmatrix} - y = \underbrace{\begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{\hat{y}^{(0)}} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 5 \\ 12 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$\hat{y}^{(0)} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}, \quad J^{(0)} = \frac{1}{2}(1^2 + 11^2) = 61$$

$$\frac{\partial J}{\partial e} = \frac{2}{2} \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$\nabla_m J = \frac{2}{2} X^T e^{(0)} = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 4 \cdot 11 \\ 3 \cdot 1 + 10 \cdot 11 \end{bmatrix} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$\frac{\partial J}{\partial b} = \frac{2}{2}(1 + 11) = 12$$

$$m^{(1)} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 0.01 \begin{bmatrix} 45 \\ 113 \end{bmatrix} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$$

$$b^{(1)} = 1 - 0.01(12) = 0.88$$

$$X_{NN}^{(1)} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} = \begin{bmatrix} 1.16 \\ 2.90 \end{bmatrix}$$

$$e^{(1)} = \hat{y}^{(1)} - y = \begin{bmatrix} 1.16 \\ 2.90 \end{bmatrix} + 0.88 \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix}$$

$$\bar{J}^{(1)} = \frac{1}{M} \sum_{j=1}^{M} e_j^2 = \frac{1}{2} \left[(-2.96)^2 + (-2.22)^2 \right] = 6.485$$

$$\bar{J}^{(1)} = 6.485$$

$m^1 \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$ $b^1 = 0.88$ $x \cdot \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$ $y = \begin{pmatrix} 5 \\ 6 \end{pmatrix}$

$\hat{y} = m_1 x_1 + m_2 x_2 + b$

$\hat{y}_1^1 = ((-1.45 \times 1) + (0.87 \times 3) + 0.88$

$= -1.45 + 2.61 + 0.88$

$= 2.04$

When $x = (1, 3)$

$\hat{y}_2^1 = (-1.45 \times 4) + (0.87 \times 10) + 0.88$

$= -5.8 + 8.7 + 0.88$

$= 3.78$

When $x = (4, 10)$ so $\hat{y}^1 = \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix}$

$e = \hat{y} - y$ ....

$e = \begin{pmatrix} 2.04 \\ 3.78 \end{pmatrix} - \begin{pmatrix} 5 \\ 6 \end{pmatrix} = \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix}$

The Gradient

$\nabla m^J = 2/n \, X^T e$

$n = 2$ so $2/n = 1$

$= \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix} = \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix}$

$1(-2.96) + 4(-2.22) = -11.88$

$3(-2.96) + 10(-2.22) = -31.08$

$\frac{\partial J}{\partial b} = 1 \Sigma e^1 = (-2.96 - 2.22)$

$= -5.18$

$m^2 = m^1 - \alpha \nabla m^J$

take $\alpha = 0.01$

$= \begin{pmatrix} -1.45 \\ 0.87 \end{pmatrix} - 0.01 \begin{pmatrix} -11.88 \\ -31.08 \end{pmatrix}$

$m_1^2 = -1.45 - 0.01(-11.88) = -1.3316$

$m_2^2 = 0.87 - 0.01(-31.08) = 1.1808$

updated biase

$b^2 = b^1 - \alpha \frac{\partial J}{\partial b}$

$b^2 = 0.88 - 0.01(-5.18)$

$= 0.88 + 0.0518$

$= 0.9318$

$m_2 = \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix}$, $b_2 = 0.9318$

$\hat{y}_1^2 = 1(-1.3316) + 1.1808(3) + 0.9318$

$= 3.1420$

$\hat{y}_2^2 = (-1.3361)(4) + 1.1808(10) + 0.9318$

$= 7.4134$

$y = \begin{pmatrix} 5 \\ 6 \end{pmatrix}$

$MSE = \frac{1}{2} \sum_{i=1}^{2} (\hat{y}_i - y_i)^2$

$= \frac{1}{2} \left[(3.1420 - 5)^2 + (7.4134 - 6)^2 \right]$

$= \frac{1}{2} \left[(-1.8574)^2 + (1.4136)^2 \right]$

$= 2.7236$

Iteration 1 - Becky

NO.

Date :

Iteration Two with Adam

Step 1 : prediction $\hat{y} = xm + b$

$$m^2 = \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix}, \quad x = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}, \quad b = 0.9318$$

$$\hat{y}_1^2 = (-1.3316)(1) + 1.1808(3) + 0.9318 = 3.1426$$

$$\hat{y}_2^2 = (-1.3316)(4) + 1.1808(10) + 0.9318 = 7.4134$$

$$\hat{y}^2 = \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix}$$

Step 2 : Errors

$$e^2 = \hat{y}^2 - y, \quad y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$e^2 = \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix}$$

Step 3 : The Gradients

A- For weights $\nabla_m J = \frac{2}{n} X^T e$

$n = 2$ so $\frac{2}{n} = \frac{2}{2} = 1$

$$\nabla_m J = X^T e \qquad \nabla_m J = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix}$$

$$= \begin{bmatrix} -1.8574 + 4(1.4134) \\ 3(-1.8574 + 10(1.4134) \end{bmatrix}$$

$$\nabla_m J = \begin{bmatrix} 3.7962 \\ 8.5618 \end{bmatrix}$$

No. Date:

B- For bias

$$\frac{\partial J}{\partial b} = \frac{2}{n}\sum_{i=1}^{n} ei \qquad = \frac{\partial \bar{J}}{\partial b} = \frac{2}{n}\sum ei$$

$$= (-1.8574) + (1.4134) = 0.$$

$$\frac{\partial \bar{J}}{\partial b} = -0.444$$

Step 4: Update weights and bias

Using $\alpha = 0.01$

A- weights $\quad m^3 = m^2 - \alpha \nabla m \bar{J}$

$$= \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix} - 0.01 \begin{bmatrix} 3.7962 \\ 8.5618 \end{bmatrix}$$

$$m^3 = \begin{bmatrix} -1.369562 \\ 1.095182 \end{bmatrix}$$

B- Bias $\quad b^3 = b^2 - \alpha \frac{\partial \bar{J}}{\partial b}$

$$= 0.9318 - 0.01(-0.444)$$

$$b^3 = 0.936424$$

Step 5: Verify

$$MSE = \frac{1}{n}\sum_{i=1}^{n} (\hat{y}_i - y_i)^2$$

$$= \frac{1}{2}\left[(3.1426 - 5)^2 + (7.4134 - 6)^2\right]$$

$$= 2.7238$$

Reference: Strang G. (2016). Introduction to Linear Algebra (5th ed.). Wellesley-Cambridge Press.